

Supplemental Material for “Antiferromagnetism induced by Peierls electron-phonon-coupling”

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A. The method of projector quantum Monte Carlo

We employ the method of projector QMC to investigate the ground-state properties of the SSH optical phonon model at finite frequency and the effective Hamiltonian in the anti-adiabatic limit, as described in Eq. (1) and Eq. (2) in the main text accordingly. The algorithm is based on the principle that the ground-state expectation value of operator $\hat{O}$ in the exact ground state $|\psi_G\rangle$ can be computed exactly via projecting a trial wave function $|\psi_T\rangle$ along the imaginary time axis

$$\langle \hat{O} \rangle = \frac{\langle \psi_G | \hat{O} | \psi_G \rangle}{\langle \psi_G | \psi_G \rangle} = \lim_{\Theta \rightarrow \infty} \frac{\langle \psi_T | e^{-\Theta H} \hat{O} e^{-\Theta H} | \psi_T \rangle}{\langle \psi_T | e^{-2\Theta H} | \psi_T \rangle} \quad (S1)$$

as long as the trial wave function $|\psi_T\rangle$ has a finite overlap with the true ground state $|\psi_G\rangle$. The algorithm is intrinsically unbiased against the choice of $|\psi_T\rangle$ assuming that the trial wave function is not orthogonal to the true ground state, namely $\langle \psi_T | \psi_G \rangle \neq 0$, which is generically satisfied for a quantum many-body system with finite size. In this paper, we choose $|\psi_T\rangle$ to be the ground-state wave function of the non-interacting part of the model under consideration. In practical QMC simulations, we set the projection parameter Θ to a finite but sufficiently large value so that the expectation value $\langle \hat{O} \rangle$ converges when larger Θ is considered. We set $\Theta = 34/t, 36/t, 38/t, 46/t, 50/t$ for $L = 6, 8, 10, 12, 14$ accordingly, each of which has been checked to be large enough for convergence. As an example, we show the AF/pseudospin-AF correlation as the function of Θ for fixed $L = 8, \omega/t = 3.0, \lambda = 0.25, U = 0$, as shown in Fig. S1. It clearly indicates that convergence of the algorithm is met when $\Theta > 20/t$.

Similar to finite temperature algorithm, Trotter decomposition is implemented by discretizing Θ in spacing $\Delta\tau = \Theta/L\tau$. The Trotter error scales as $\Delta\tau^2$. In this work we choose $\Delta\tau = 0.1/t$. The convergence of the discretization has also been checked by comparing results with smaller $\Delta\tau$.

The SSH model (Eq. (1) in the main text) becomes quadratic in electron operators for a specific space-time phonon configuration. We compute the electron's Green's function for each phonon configuration, and sample the phonon configuration by Monte Carlo. As for the effective model in the anti-adiabatic limit (Eq. (2) in the main text), auxiliary fields are introduced into the Hamiltonian via a SU(2) symmetric Hubbard-Stratonovich decomposition. The AF and VBS order parameters for a finite system size $N = L^2$ are given by

$$\hat{O}_{\text{AF}}(\mathbf{q}) = \frac{1}{L^2} \sum_j e^{i\mathbf{q} \cdot \mathbf{R}_j} \hat{S}_j^z \quad (S2)$$

$$\hat{O}_{\text{VBS}}(\mathbf{q}) = \frac{1}{L^2} \sum_j e^{i\mathbf{q} \cdot \mathbf{R}_j} \left(\hat{B}_{j,\hat{x}} + i\hat{B}_{j,\hat{y}} \right) \quad (S3)$$

where $\hat{B}_{j,\delta} = c_{j,\sigma}^\dagger c_{j+\delta,\sigma} + h.c.$ is the kinetic operator on $\delta = \hat{x}, \hat{y}$ bonds. VBS breaks the lattice $\mathbb{Z}_4$ symmetry while AF breaks spin SU(2) rotational symmetry. The structure factors for each order parameter with momentum $\mathbf{Q}$ is defined as $S(\mathbf{Q}) = \langle |\hat{O}(\mathbf{Q})|^2 \rangle$. For both AF and staggered VBS, the peaked momentum is $\mathbf{Q} = (\pi, \pi)$. The AF susceptibility can be computed as an integration over the imaginary-time interval τ_M

$$\chi_{\text{AFM}}(\mathbf{q}) = \int_{\Theta - \tau_M/2}^{\Theta + \tau_M/2} d\tau \langle \hat{O}_{\text{AFM}}(\mathbf{q}, \tau) \hat{O}_{\text{AFM}}(\mathbf{q}, 0) \rangle. \quad (S4)$$

To compute $\chi_{\text{AF}}(\mathbf{q})$, the imaginary-time interval τ_M should be large enough such that χ is converged, while the correlators $\langle \hat{O}(\mathbf{q}, \tau) \hat{O}(\mathbf{q}, 0) \rangle$ should be computed after sufficiently long projection. Thus, in practice we choose $\tau_M \approx \Theta/4$ and integrate over the correlators from $\Theta - \tau_M/2$ to $\Theta + \tau_M/2$.

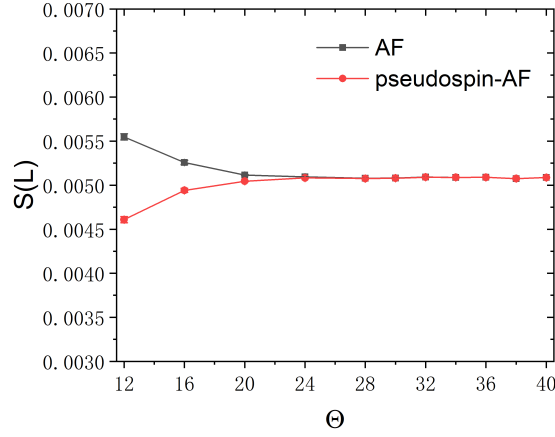


FIG. S1. Convergence check for $L = 8, \omega/t = 3.0, \lambda = 0.25$. QMC results of AF/pseudospin-AF correlation as a function of Θ .

B. Derivation of the effective Hamiltonian in the anti-adiabatic limit

Here we derive the effective phonon-mediated interaction the anti-adiabatic (AA) limit described by Eq. (2) of the main text. The imaginary-time action is given by

$$S[\bar{\psi}, \psi, X] = \int d\tau \sum_{\langle ij \rangle} \left[\bar{\psi} \partial_\tau \psi - t B_{ij} + \frac{M}{2} (\partial_\tau X_{ij})^2 + \frac{K}{2} X_{ij}^2 + g X_{ij} B_{ij} \right] \quad (\text{S5})$$

where $B_{ij}(\tau) = \sum_\sigma \bar{\psi}_{i\sigma} \psi_{j\sigma} + \text{h.c.}$ is the electronic hopping on bond $\langle ij \rangle$. In AA limit $M \rightarrow 0$ (or $\omega \rightarrow \infty$), the phonon kinetics approaches to zero and the phonon fields X_{ij} can be integrated out directly to generate the following effective action:

$$S_{\text{eff}}[\bar{\psi}, \psi] = \int d\tau \sum_{\langle ij \rangle} \left[\bar{\psi} \partial_\tau \psi - t B_{ij} - \frac{g^2}{2K} B_{ij}^2 \right]. \quad (\text{S6})$$

By rewriting B_{ij}^2 in terms of spin and pseudospin operators, one finds that the effective Hamiltonian is given by:

$$H_{\text{AA}} = -t \sum_{\langle ij \rangle \sigma} \left(c_{i\sigma}^\dagger c_{j\sigma} + \text{h.c.} \right) + J \sum_{\langle ij \rangle} \left[\mathbf{S}_i \cdot \mathbf{S}_j + \tilde{\mathbf{S}}_i \cdot \tilde{\mathbf{S}}_j \right] \quad (\text{S7})$$

where $J = g^2/K$. The effective interaction is the summation of spin and pseudospin AF exchange coupling, namely the terms in Eq. (2) of the main text.

C. Notes on finite-size analysis

In the main text, we performed the finite-size scaling to obtain the quantities in the thermodynamic limit. The scaling function we usually use is $C(L) = C(L \rightarrow \infty) + a/L^b$, where $C(L)$ stands for the quantity such as AF order parameter and spin gap computed for system size L . We estimate the order parameter in finite lattice as the square root of structure factor $M_{\text{AF}} = \sqrt{3 S_{\text{AF}}(\pi, \pi)}$, where factor 3 comes from the equal contribution of x, y, z components of the spin structure factor. In Fig. 2(b) in the main text, we presented the result of AF order parameters as a function of λ in the anti-adiabatic limit. For $\lambda \approx 0.25$ and finite phonon frequency ($0 < \omega < \infty$), the extrapolated AF order as a function of ω is shown in Fig. S2. The value at $t/\omega = 0$ comes from the computation at anti-adiabatic limit. The critical frequency ω_c discussed in the main text is the phonon frequency where M_{AF} vanishes.

We compute spin gap Δ_s according to the asymptotic scaling behavior of time-dependent spin-spin correlation: $\langle S^z(\mathbf{Q}, \tau) S^z(\mathbf{Q}, 0) \rangle \sim e^{-\Delta_s \tau}$, where $\mathbf{Q} = (\pi, \pi)$. The spin gap $\Delta_s(L)$ for different lattice size L can be extracted by fitting the correlation when τ is sufficiently large. Then, $\Delta(L)$ is extrapolated to $L \rightarrow \infty$ via a power-law scaling function $\Delta(L) = \Delta + a/L^b$ to obtain the spin gap in the thermodynamic limit.

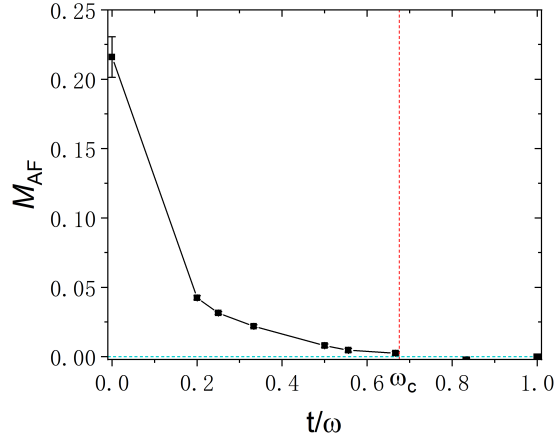


FIG. S2. The QMC-computed AF order parameter as a function of inverse phonon frequency at $\lambda \approx 0.25$ ($g = 1.4$). The red dashed line represents value of corresponding critical frequency discussed in the main text.

D. Exact solution in the adiabatic limit

In the adiabatic limit ($\omega = 0$), phonon displacements X_{ij} become classical quantities without quantum dynamics. At zero-temperature, the phonon displacements are static and the fermions are described by a quadratic Hamiltonian depending on the phonon displacement $X = \{X_{ij}\}$. Consequently, an exact solution of the ground state of the SSH model in the adiabatic limit can be obtained in the variational sense. In the adiabatic limit, SSH model is reduced to the following quadratic form depending on the phonon configuration X :

$$H[X] = \sum_{i,\delta} \left[\frac{K}{2} X_{i,\delta}^2 + (gX_{i,\delta} - t) (c_{i,\sigma}^\dagger c_{i+\delta,\sigma} + h.c.) \right], \quad (\text{S8})$$

where $\delta = x, y$ refers to unit vectors on x or y directions and $X_{i,\delta} = X_{i,i+\delta}$ is the phonon displacement on the NN bond $\langle i, i + \delta \rangle$. The variational method is applied here by minimizing the ground-state energy, which yields the following self-consistent equation: $g \langle (c_{i,\sigma}^\dagger c_{i+\delta,\sigma} + h.c.) \rangle = -KX_{i,\delta}$.

Here we compute the energy assuming various different ansatz of the phonon displacement configuration X , including staggered, staircase, columnar, and plaquette VBS patterns. The ordering vector for staggered and staircase patterns is (π, π) while the ordering vector of columnar and plaquette patterns is $(\pi, 0)$ or $(0, \pi)$. Since the Fermi surface of the non-interacting electrons at half filling is perfectly nested by the wave vector (π, π) , it is expected that the VBS patterns with ordering vector (π, π) is more favored than the ones with $(0, \pi)$ or $(\pi, 0)$. Both the staggered and columnar VBS patterns break the lattice $\mathbb{C}_4$ symmetry while the staircase and plaquette ones develop dimer ordering on both x and y directions. We assume $X_{i,\delta} = m_\delta + (-1)^{i_x+i_y} \Delta_\delta^{\text{stag}} + (-1)^{i_\delta} \Delta_\delta^{\text{col}}$, where m_δ is the uniform component of the phonon displacements such that the uniform hopping amplitude along $\delta = x, y$ direction is given by $t_\delta = t - gm_\delta$. Our calculations clearly show that the SSH phonons in the adiabatic limit favors the staggered VBS state with momentum (π, π) and $\mathbb{C}_4$ symmetry breaking for any finite EPC constant λ . Besides, when the staggered VBS order parameter develops spontaneously in x -direction, namely $\Delta_x^{\text{stag}} \neq 0$ while $\Delta_y^{\text{stag}} = 0$, the uniform hopping amplitude t_x and t_y exhibits the anisotropy $t_x > t_y$. Such anisotropy develops so that the shifted Fermi surface $-2t_x \cos k_x - 2t_y \cos k_y = 0$ does not cross the nodal lines of gap function $2\Delta_x^{\text{stag}} \sin k_x = 0$; namely the Fermi surface can be fully gapped to gain energy.

The staggered VBS is expected to be stable against perturbations that break Fermi surface nesting such as small next nearest-neighbor hopping t' , particularly when the electron-phonon coupling is strong. However, if t' is large enough, pairing instabilities might emerge and superconductivity is expected to be dominant. When the t' is large and Fermi surface is far from nesting, it is also possible that the VBS order with different pattern emerges. The SSH model without Fermi surface nesting is interesting and deserves further studies in future.

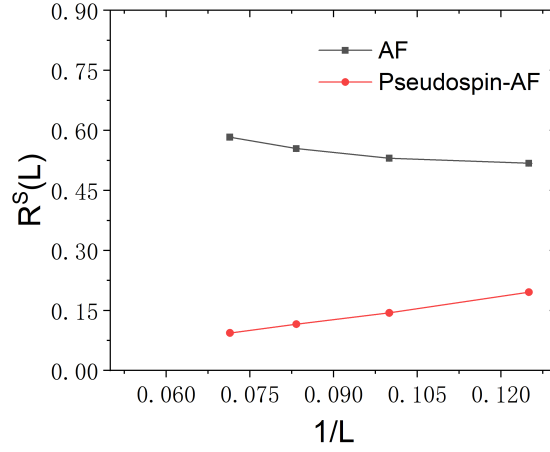


FIG. S3. The finite size scaling of correlation ratio of AF and pseudospin-AF ordering at $\omega = 3t$ and $\lambda \approx 0.25$ ($g = 1.4t$) for $U = 0.5$.

E. The effect of repulsive Hubbard interaction

As mentioned in the main text, the SSH model respects $O(4)$ symmetry, which implies the degeneracy between AF and pseudospin-AF (namely CDW/SC) correlations. However, this degeneracy can be lifted by any finite (even infinitesimal) Hubbard interaction. By turning on a weak repulsive Hubbard interaction H_U , AF is more favored than pseudospin-AF ordering so that the degeneracy between them is lifted. For instance, in Fig. S3 we present the results of turning on a weak Hubbard U at $\omega = 3t$ and $\lambda \approx 0.25$ ($g = 1.4t$) where the ground state without U is degenerate between AF and pseudospin-AF (namely CDW/SC) ordering. Once a weak Hubbard $U = 0.5$ is turned on, pseudospin-AF (CDW/SC) order is suppressed to zero and AF is the only order in the ground state, as shown in Fig. S3. Thus, in the presence of any weak repulsive Hubbard interaction, AF is the only order when the phonon frequency is larger than a critical value or the EPC constant is less than a critical value.

F. The inter-correlation between spin and pseudospin

As we discussed in the main text, the degeneracy between the states with spin AF or pseudospin AF order for $U = 0$ does not imply their coexistence in the ground state. To provide direct evidence on whether the spin and pseudospin AF orders coexist in the ground state, we calculate the inter-correlation between the two orders, namely

$$C = \left(\frac{1}{L^2}\right)^4 \sum_{i,j,k,l} \left\langle [(-1)^{i-j} \mathbf{S}_i \cdot \mathbf{S}_j] [(-1)^{k-l} \tilde{\mathbf{S}}_k \cdot \tilde{\mathbf{S}}_l] \right\rangle. \quad (\text{S9})$$

For simplicity, we compute the inter-correlations between the z components of spin and pseudospin operators. If the two orders coexist, the inter-correlation should be non-zero in the thermodynamic limit; otherwise this inter-correlation should be extrapolated to zero. The finite size scaling results of inter-correlation have already been shown in the main text. The results clearly indicate that the inter-correlation extrapolates to zero for $U = 0$ in the thermodynamic limit. As no correlation between the two orders can be detected, QMC simulation samples the ground-state phase space in either spin AF sector or pseudospin AF sector, implying that the two orders do not co-exist in the ground state. The results unambiguously confirm that particle-hole $\mathbb{Z}_2$ symmetry of spin-down electrons is spontaneously broken, and spin and pseudospin AF ordering does not coexist in the ground state for $U = 0$.

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